5.

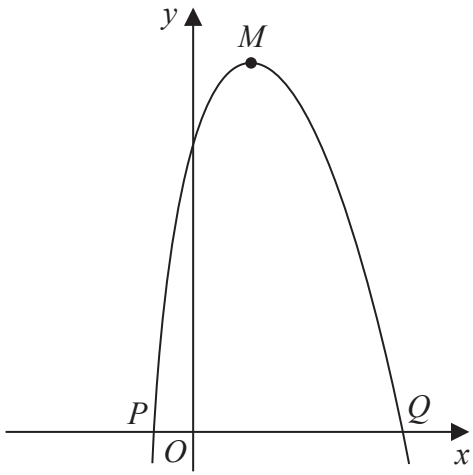



**Figure 1**

Figure 1 shows a sketch of part of the curve with equation

$$y = 6\ln(2x + 3) - \frac{1}{2}x^2 + 4 \qquad x > -\frac{3}{2}$$

The curve cuts the negative $x$-axis at the point $P$, as shown in Figure 1.

(a) Show that the $x$ coordinate of $P$ lies in the interval $[-1.25, -1.2]$

**(2)**

The curve cuts the positive $x$-axis at the point $Q$, also shown in Figure 1.

Using the iterative formula

$$x_{n+1} = \sqrt{12\ln(2x_n + 3) + 8} \quad \text{with} \quad x_1 = 6$$

(b) (i) find, to 4 decimal places, the value of $x_2$

(ii) find, by continued iteration, the $x$ coordinate of $Q$. Give your answer to 4 decimal places.

**(3)**

The curve has a maximum turning point at $M$, as shown in Figure 1.

(c) Using calculus and showing each stage of your working, find the $x$ coordinate of $M$.

**(4)**